# What Light Is, and How Fast


SYNOPSIS

Light turns out to be a solution of Maxwell's equations — an electromagnetic wave that carries itself along and needs nothing to travel through. A speed falls straight out of those equations, and it matched a number experimenters had already measured. This note is that story, and then what becomes of that speed when the light is in water or glass instead of nothing.


BEFORE YOU START — PREDICT

Light crosses empty space at $c = 3\times10^8$ m/s. Now send it into a block of glass. **At what speed, compared with *c*, does it travel there** — faster, slower, or the same? Write down your answer and one sentence of reason. Section 2 settles it.

PERFORMANCE EXPECTATIONS

1. **Say** what light is, in the sense Maxwell's equations give: an electromagnetic wave, self-sustaining, needing no medium.
2. **State** the relationship between a changing electric field and a changing magnetic field, and say which causes which.
3. **Explain** why $c = 1/\sqrt{(\varepsilon_0\mu_0)}$ was the clue that light is electromagnetic, given that light's speed had already been measured.
4. **Use** $v = c/n$ to find the speed of light in a transparent medium, and say why $n > 1$ means what it means.

## 1. What is light?

To this day, one can't say light is *fully* understood — that story runs on into quantum mechanics. But the clues as to its nature were first gleaned from the work on electricity and magnetism by Faraday, and especially by Maxwell.

Faraday's experimental results, and his idea of **fields** permeating and propagating through space and time, were formulated rigorously by Maxwell in his famous equations. **Light is a solution of Maxwell's equations**: an electromagnetic wave travelling at a fixed speed,

$$c = 1 / \sqrt{(\varepsilon_0\mu_0)} = 3\times10^8 \text{ m/s}$$

Counter-intuitively, such a disturbance of electromagnetic nature is **self-sustaining**, requiring no medium. Light propagates through vacuum at speed $c$, the maximal speed.

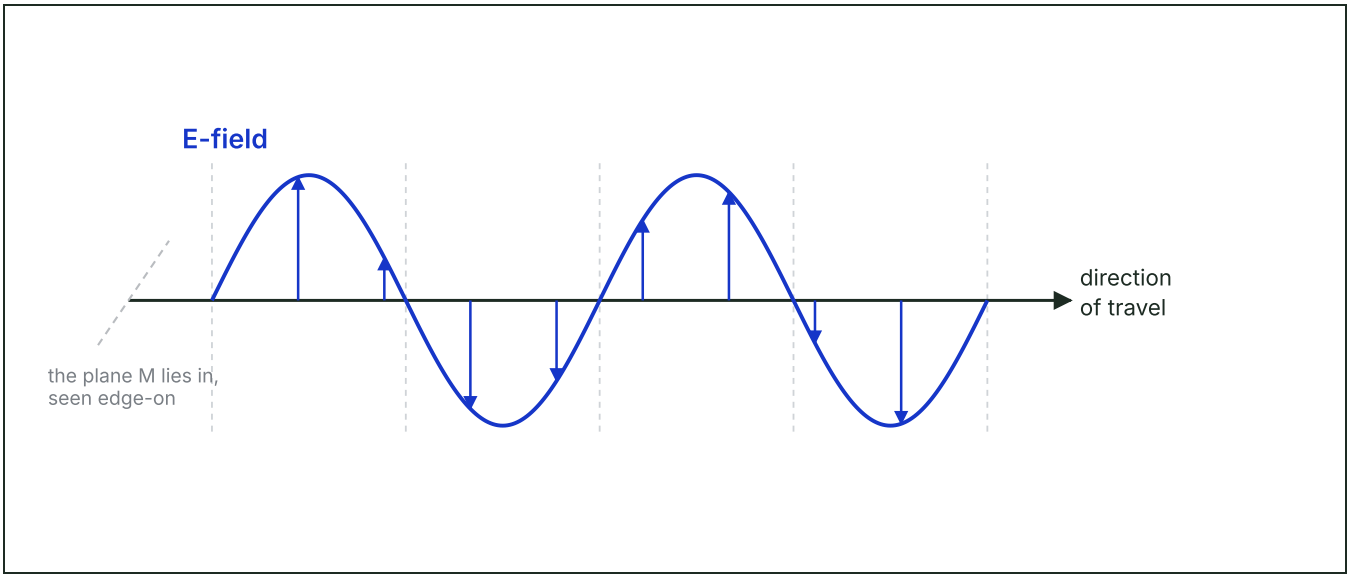



**Figure 1.** The two fields rise and fall *together*: at every dashed line both are zero at once, and between them both are at full strength at once. M is drawn receding, in the plane at right angles to E — which is why its crest appears shifted to the right on the page even though it sits at the same point along the axis. Nothing is oscillating up and down *in* the wave: no rope, no air, no medium at all. What travels is the pattern of the fields themselves.

**Your job on Figure 1.** The electric field is drawn for you. Add the magnetic field: same wavelength, same zero crossings, peaking where E peaks — but in the plane shown edge-on at the left, not in E's plane.

**KEY**

Change (w.r.t. time) in $E$ causes change (w.r.t. time) in $M$

That is the engine of the whole thing. A changing electric field makes a changing magnetic field, which makes a changing electric field, and so on down the line — which is why the wave keeps going with nothing to travel through.

## 2. How fast does light travel?

Maxwell's equations, which were written to encapsulate the known experimental results of Faraday and Ampere and others, not only reveal that light is an electromagnetic wave — they reveal **how fast it should travel**.

Light speed was measured *before* anyone thought light was an electromagnetic wave. The electromagnetic nature of light was inferred from the fact that **$1/\sqrt{(\varepsilon_0\mu_0)}$ happened to match the then-known value of the speed of light**.

Light speed $c$ is the speed when light travels in **vacuum** — that is, in nothing. But light can also travel through transparent media such as water or glass.

1. At what speed, compared with $c$, will light travel in a transparent medium?

THE PUNCHLINE

$$v = c/n, \text{ with } n > 1$$

One constant out of two constants of electromagnetism, and then one number per material that says how much the material slows the light down. That single number $n$ is what the whole of the next three notes runs on.

## Looking ahead

You now have what light is and how fast it goes. **Note 2** is the **law of reflection** — what happens when that light arrives at a shiny surface instead of carrying straight on.

2. Which statement about an electromagnetic wave is correct?

○ It needs a medium to travel through, like sound does. ○ The electric and magnetic fields are parallel to each other. ○ The two fields are at right angles to each other and to the direction of travel. ○ Only the electric field oscillates; the magnetic field is constant.

3. Why was it such strong evidence that light is electromagnetic when $1/\sqrt{(\varepsilon_0\mu_0)}$ was worked out?

○ Because the speed of light had already been measured, and the two numbers agreed. ○ Because it proved light needs no medium. ○ Because $\varepsilon_0$ and $\mu_0$ are properties of light itself. ○ Because it showed light is the fastest thing there is.

4. Light slows to $2.00\times10^8$ m/s inside a certain glass. What is the refractive index of that glass?

Check Clear

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